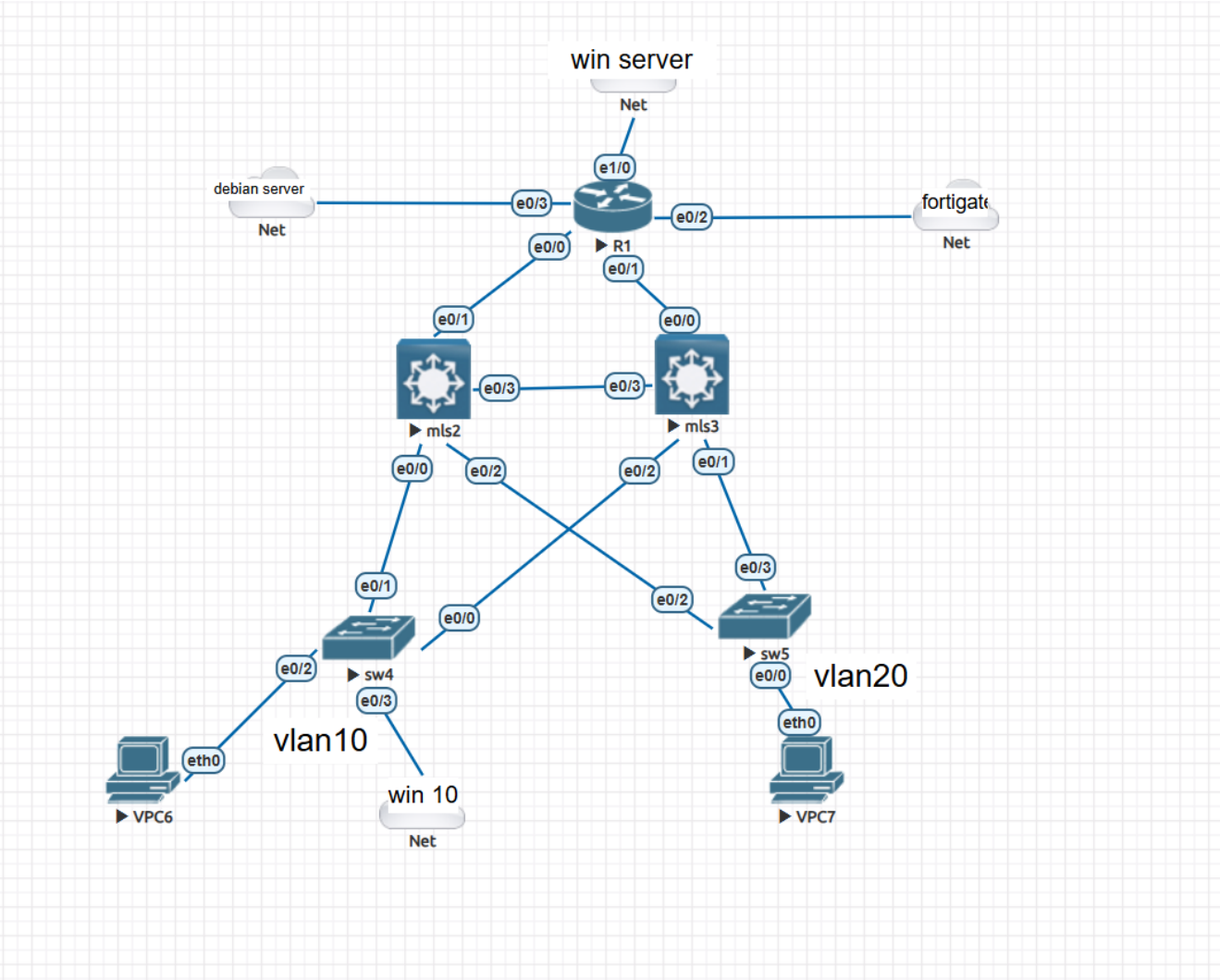
Enterprise Infrastructure Design

Multi-Vendor Integration Report

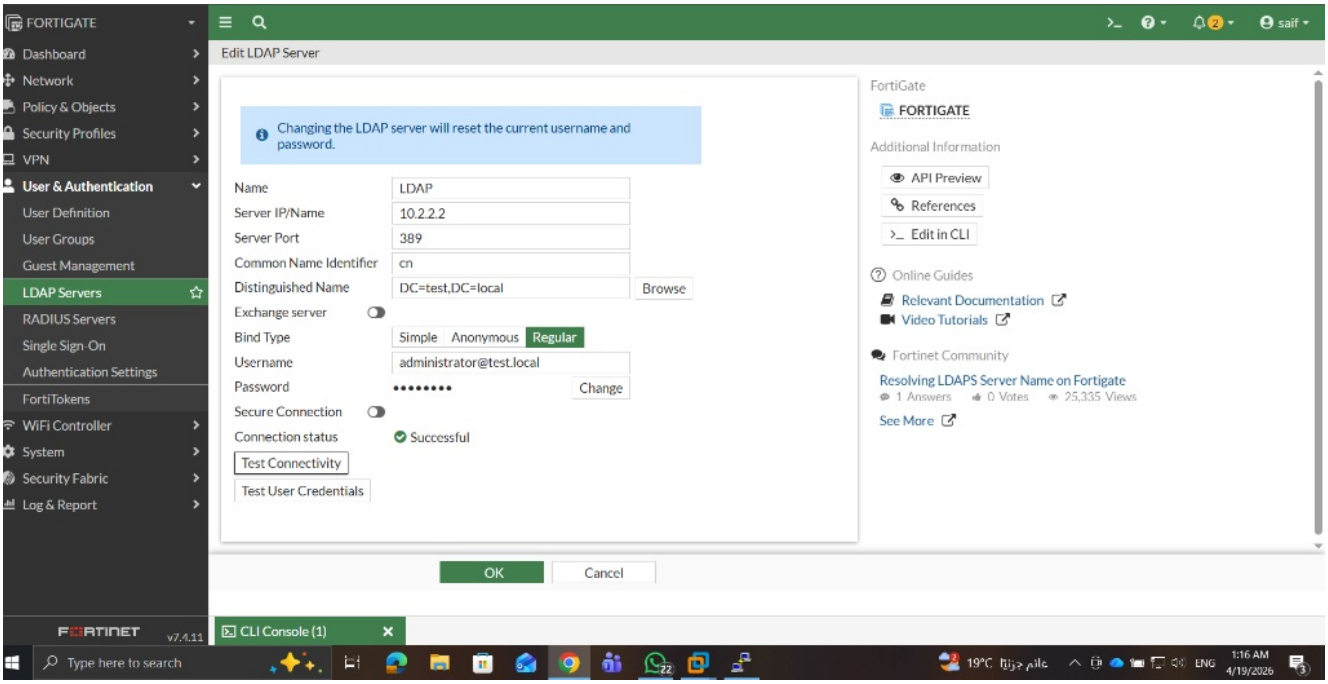
Prepared by: Seif Eldeen Elsayed

1. Enterprise Infrastructure Design: Multi-Vendor Integration



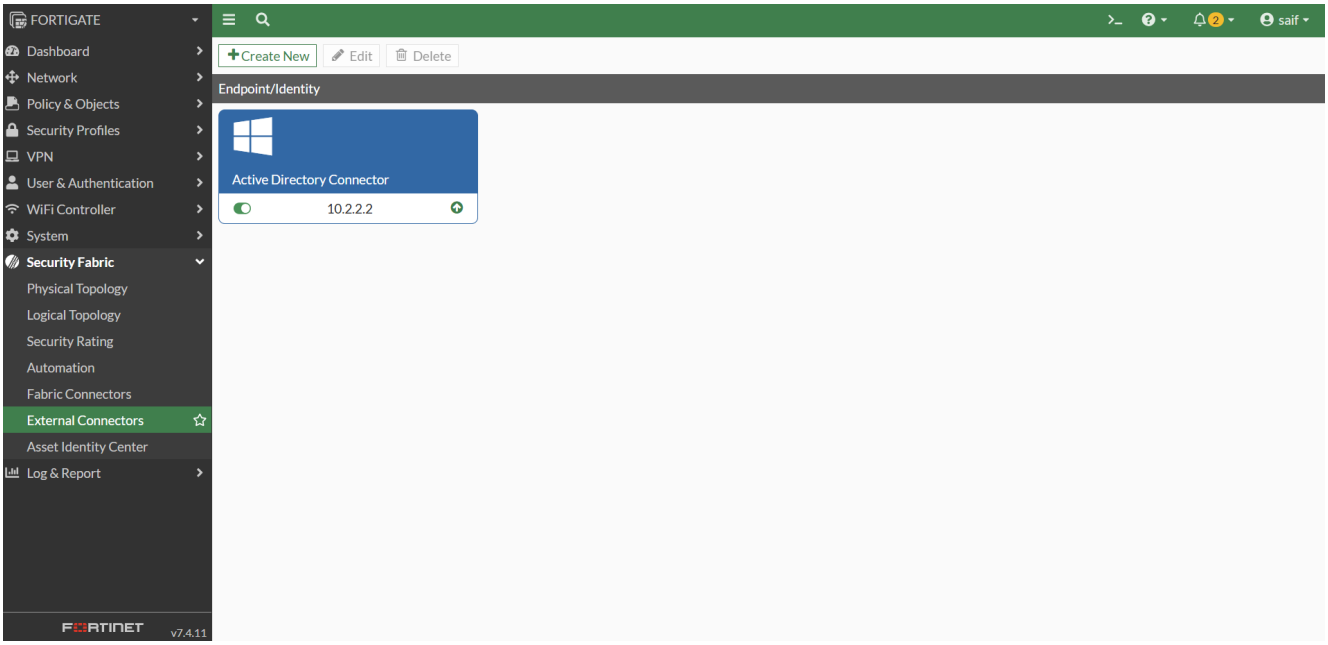
A high-level view of the simulated enterprise network featuring Cisco core switching, Fortinet security edge, and a hybrid Windows/Linux server environment.

2. Identity Integration: LDAP Bind on FortiGate



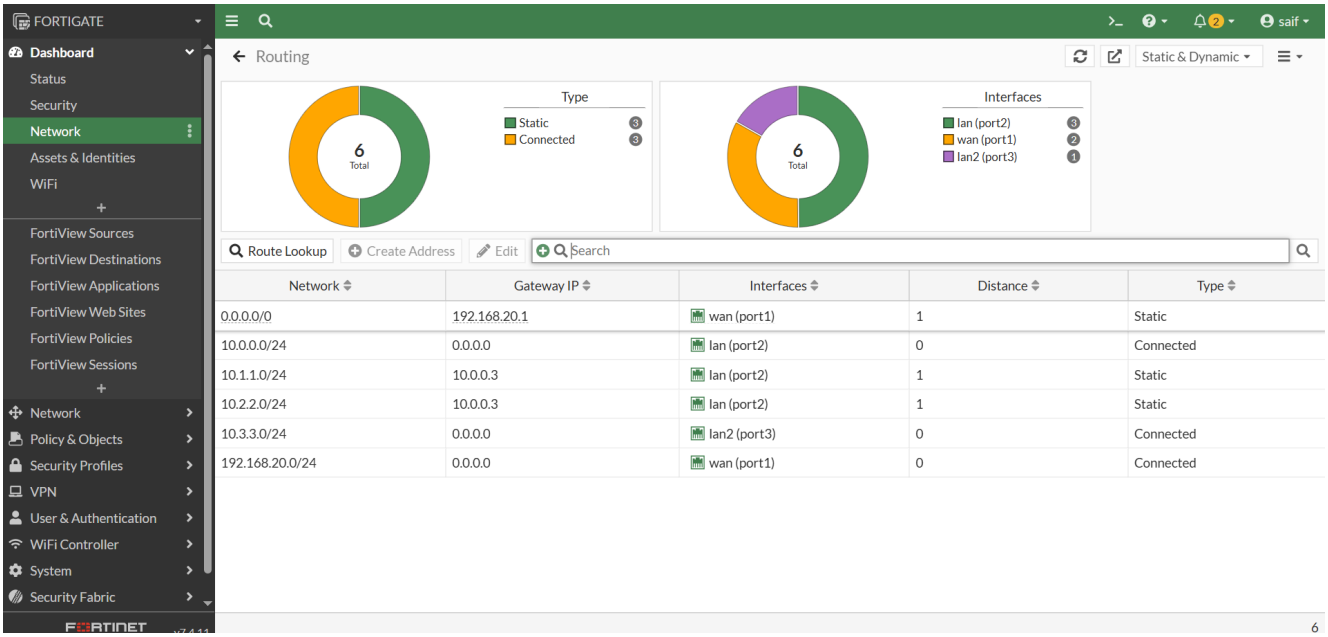
Establishing a secure LDAP connection between FortiGate and Windows Active Directory to serve as the single source of truth for user identification.

3. Transparent Access: Bypassing Captive Portals



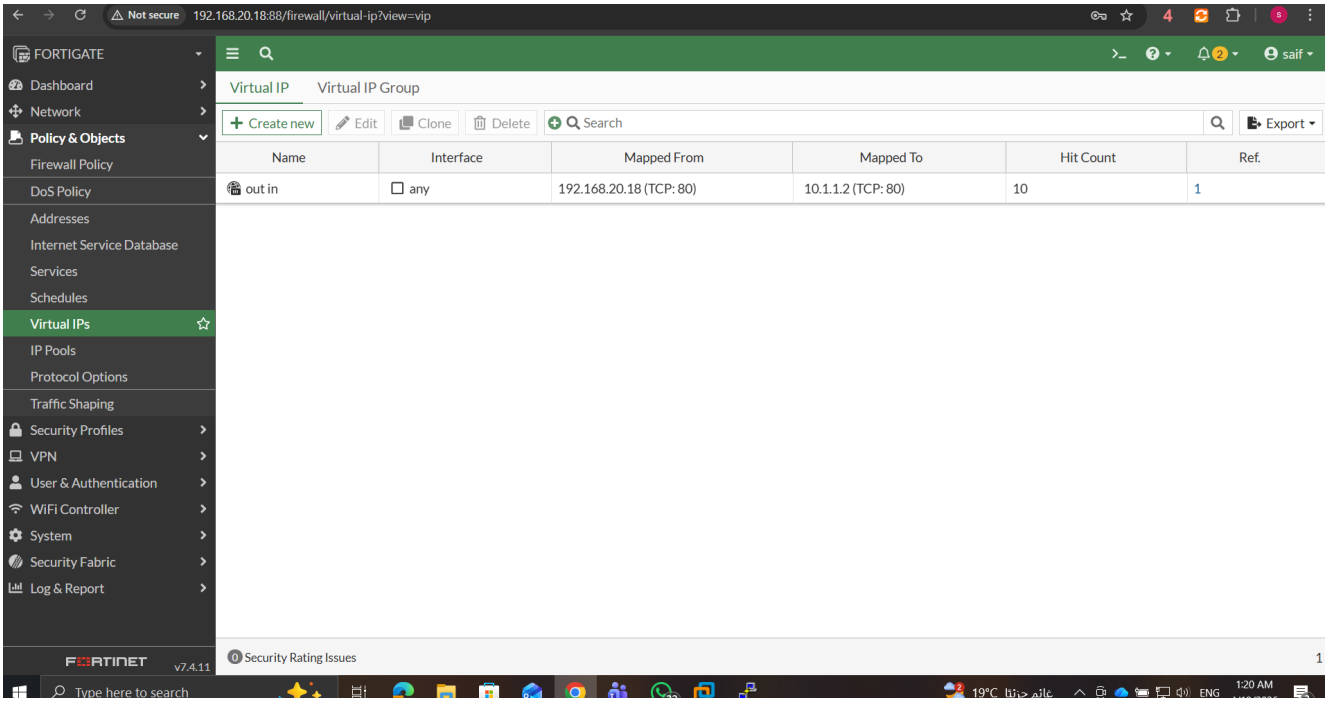
Leveraging AD Connectors to enable identity-based policies. Users are identified dynamically, granting internet access without requiring manual login portals.

4. FortiGate Static & Connected Routing



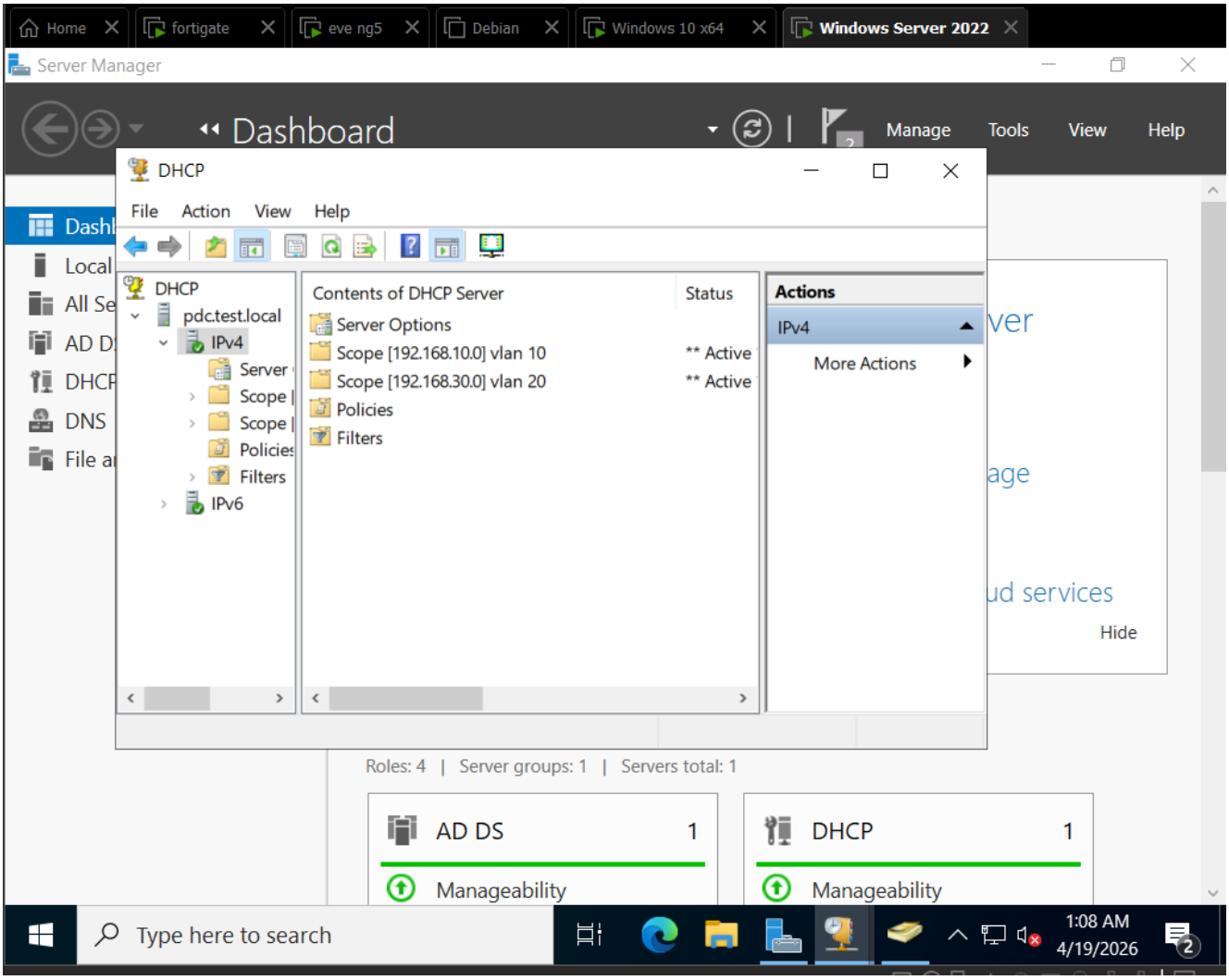
Managing the edge routing table to ensure proper traffic redirection between the internal subnets and the WAN interface.

5. Inbound Access: Destination NAT (VIP) Configuration



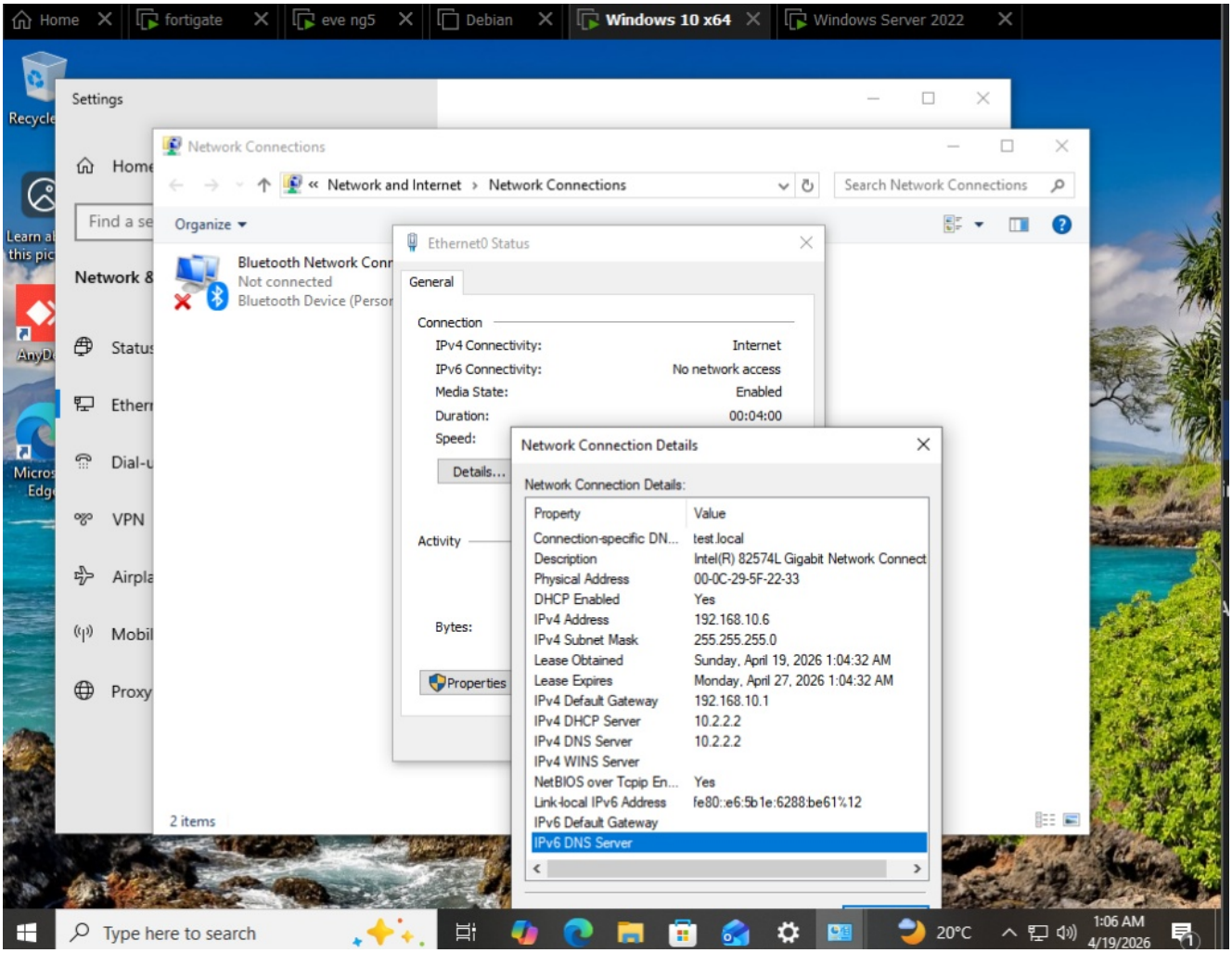
Configuring Virtual IPs (VIPs) to securely map external traffic to the internal Debian Web Server on Port 80.

6. Centralized IP Management: Windows DHCP Scopes



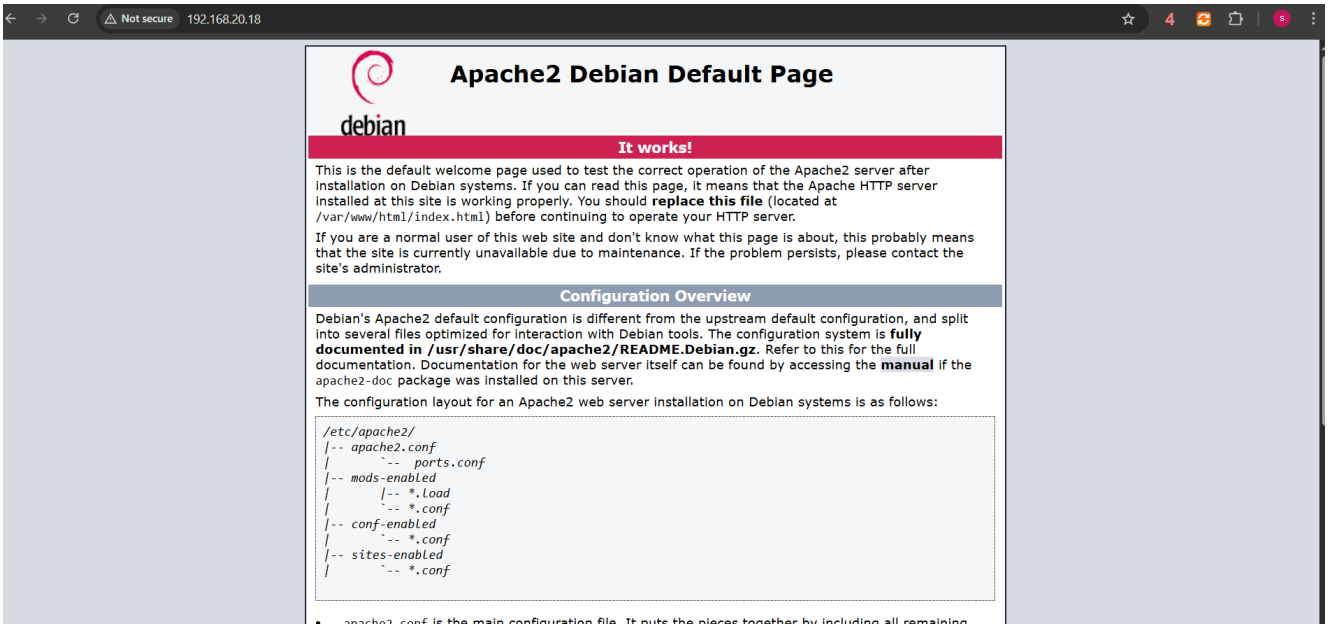
Active DHCP scopes for VLAN 10 and VLAN 20, providing automated IP addressing for all network endpoints.

7. Endpoint Connectivity: Win10 Client Validation



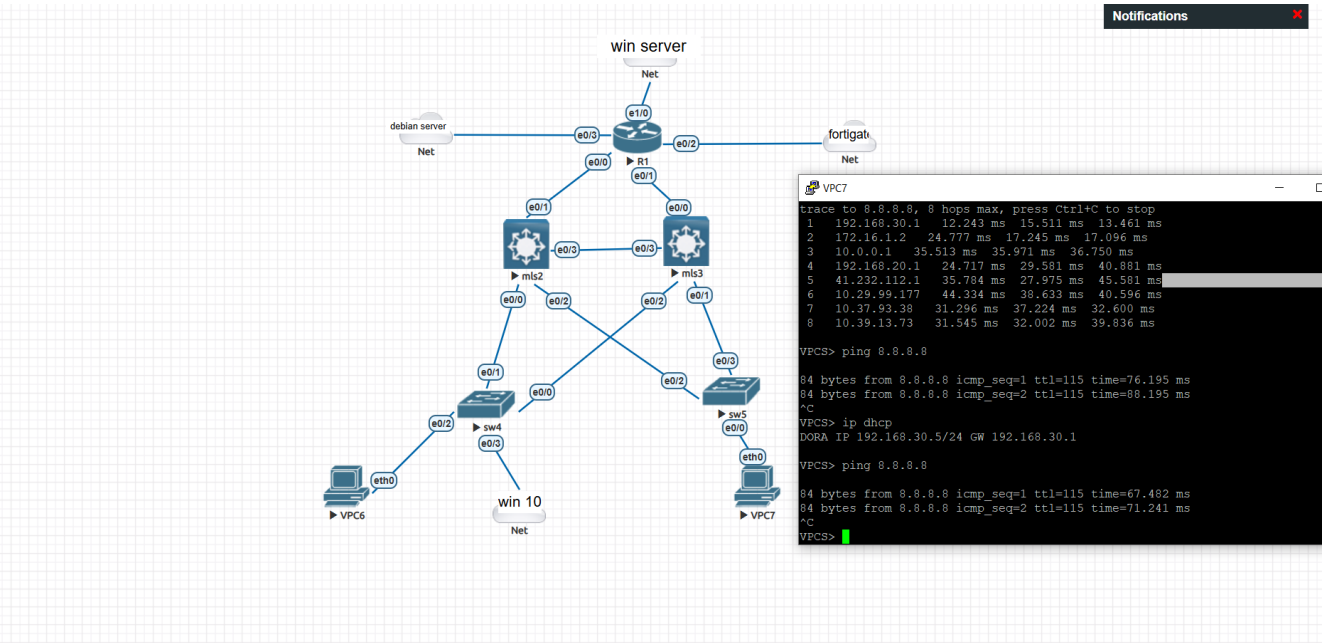
Verification of a Windows 10 host successfully obtaining an IP from the central DHCP server and confirming connectivity within the domain.

8. Application Layer: Debian-based Web Server



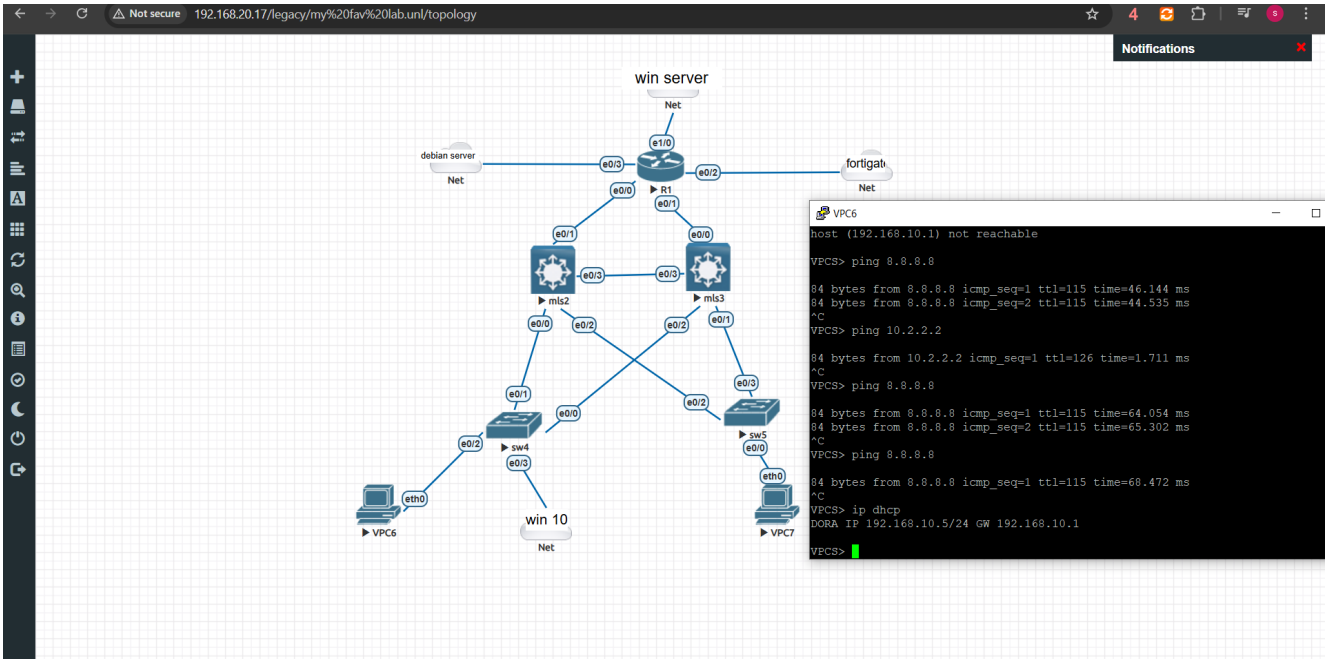
A live look at the internal Web Server (Apache2) running on Debian, serving content to authenticated users within the network.

9. Testing & Validation: VPC7 Trace Route



Successful trace route and ICMP testing from VLAN 20, confirming full reachability from the internal host to the public DNS (8.8.8.8).

10. DORA Process & Internet Access Verification



Final verification of the DORA process on VPC6, demonstrating seamless IP acquisition and internet connectivity across the Cisco/Fortinet fabric.